

MUHAMMAD HAMZA RAFI

Software Engineer

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CAREER OBJECTIVE:

Computer Science student skilled in software development with hands-on experience in HTML, C++, Python, Java, JavaScript, React and Angular. Complemented by a strong background in customer service, sales, and financial transactions, with proven strengths in communication, problem-solving, and client-focused solutions. Seeking a challenging role to apply both technical skills and interpersonal abilities in contributing to innovative, user-centered software projects.

WORK EXPERIENCE:

Air University (National Cybercrime and Forensic Lab)

Intern

July, 2025 - September, 2025

- Developed a comprehensive Browser Forensics Tool using React.js and Python (FASTAPI) to automate the extraction and visualization of critical artifacts, including history, cookies, and login credentials
- Engineered a real-time monitoring system that tracks live browsing activity and triggers instant UI notifications, significantly reducing the time required to detect and log new user behaviour.
- Designed a modular backend architecture to parse and process SQLite databases from multiple browser engines, ensuring structured data delivery via RESTful APIs for seamless frontend integration.
- Architected an interactive Timeline Builder within the UI, allowing forensic analysts to chronologically reconstruct user sessions and identify patterns of suspicious activity at a glance.
- Optimized data extraction workflows by implementing separate processing modules for different artifact types, enhancing the application's scalability and ease of maintenance for forensic investigations.
- <https://github.com/hamzee06/BrowserArtifact-TimelineBuilder>

Speridian Technologies

Intern

July, 2024 - September, 2024

- Developed and enhanced web applications using the Angular JavaScript framework, contributing to a more responsive and user friendly interface.
- Participated in the development of the "Project Management Performance" system, aimed at tracking and improving project execution efficiency.
- Collaborated closely with senior developers to design and implement both front-end and back-end components, ensuring seamless functionality
- Managed data operations using MySQL, including database design, data storage, retrieval, and performance optimization.
- Gained practical experience in web development, debugging, performance tuning, and integrating front-end with backend services, strengthening full-stack development skills.
- https://github.com/hamzee06/Performance_Measurement_Plan

RELEVANT PROJECTS:

AI-Driven Mental Health Analytics & Coaching Platform

Final Year Project

- Advanced AI Integration: Developed a specialized mental health chatbot by fine-tuning an Ollama LLM for professional communication and implementing a RoBERT-based model for highaccuracy emotion and mood detection.
- Full-Stack Clinical Dashboard: Engineered a responsive React frontend and Django backend to visualize patient sentiment trends and automate the summarization of therapist-client interactions.
- Real-Time Crisis Management: Architected a critical-alert system within the SQLite3 database structure that identifies high-risk emotional states and instantly connects clients with human coaches during emergencies.
- Performance & Fine-Tuning: Optimized model inference and data processing pipelines in Python, ensuring the AI provides contextaware responses while maintaining data privacy and integrity.

The Trinket Bloom – Online Shop

Founder & Lead Developer

- Conceptualized and launched a high-end e-commerce platform for handcrafted resin jewelry, focusing on an intuitive user journey.
- The application features a dynamic product catalog for unique items like earrings and pendants, supported by a custom-built API architecture to handle real-time data persistence and inventory management.
- By prioritizing a mobile-first responsive design and modular interface, the platform provides customers with a seamless transition from product discovery to secure checkout, effectively bridging the gap between artisanal craftsmanship and modern digital retail.
- <https://thetrinketbloom.shop/>

Portfolio

- Developed a responsive, component-based personal portfolio featuring a custom-themed UI and smooth scroll navigation.
- Implemented a dynamic animated particle background and a Dark/Light mode toggle to enhance user engagement and accessibility.
- Designed an interactive Skills Visualization module and a Projects Gallery integrated with GitHub API links for live code demonstration.
- Deployed the application via Vercel with a focus on performance optimization and mobile-first design principles.
- <https://portfolio2-six-rouge.vercel.app/>

EDUCATION:

2022 - 2026 | Foundation University Rawalpindi Campus

Bachelor of Computer Science

SKILLS:

- JavaScript (React.js, AngularJS)
- Python (Django, RESTful APIs)
- Ollama
- Transformer Model (RoBERTa)